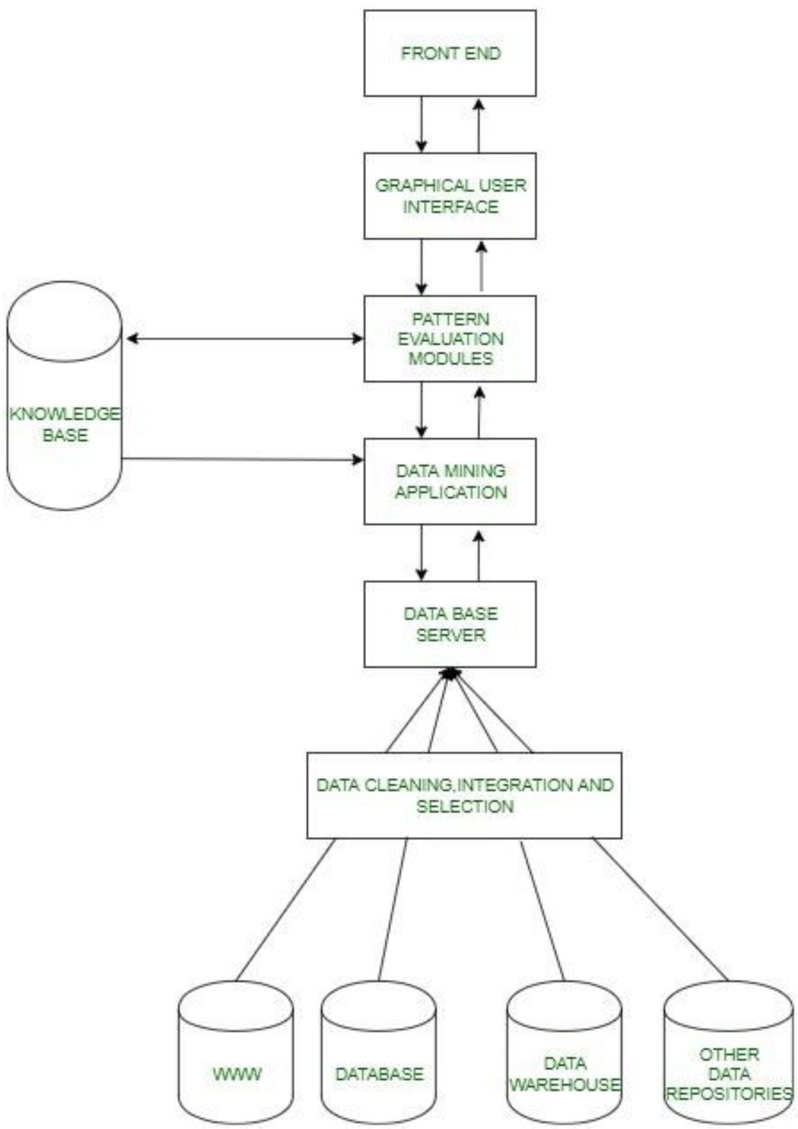
Unit III - Data Mining -II

Architecture of A Typical Data Mining Systems- Classification of Data Mining Systems. Associations and Correlations, Association Rule Mining: - Efficient and Scalable Frequent Item set Mining Methods – Mining Various Kinds of Association Rules – Association Mining to Correlation Analysis – Constraint-Based Association Mining.

Architecture of a Typical Data Mining Systems- Classification of Data Mining Systems.

Data Mining refers to the **detection and extraction** of new patterns from the already collected data. Data mining is the amalgamation of the field of statistics and computer science aiming to discover patterns in incredibly large datasets and then transform them into a comprehensible structure for later use.

The architecture of Data Mining:



Basic Working:

1. It all starts when the user puts up certain data mining requests, these requests are then sent to data mining engines for pattern evaluation.
2. These applications try to find the solution to the query using the already present database.
3. The metadata then extracted is sent for proper analysis to the data mining engine which sometimes interacts with pattern evaluation modules to determine the result.
4. This result is then sent to the front end in an easily understandable manner using a suitable interface.

A detailed description of parts of data mining architecture is shown:

1. **Data Sources:** Database, World Wide Web(WWW), and data warehouse are parts of data sources. The data in these sources may be in the form of plain text, spreadsheets, or other forms of media like photos or videos. WWW is one of the biggest sources of data.
2. **Database Server:** The database server contains the actual data ready to be processed. It performs the task of handling data retrieval as per the request of the user.
3. **Data Mining Engine:** It is one of the core components of the data mining architecture that performs all kinds of data mining techniques like association, classification, characterization, clustering, prediction, etc.
4. **Pattern Evaluation Modules:** They are responsible for finding interesting patterns in the data and sometimes they also interact with the database servers for producing the result of the user requests.
5. **Graphic User Interface:** Since the user cannot fully understand the complexity of the data mining process so graphical user interface helps the user to communicate effectively with the data mining system.
6. **Knowledge Base:** Knowledge Base is an important part of the data mining engine that is quite beneficial in guiding the search for the result patterns. Data mining engines may also sometimes get inputs from the knowledge base. This knowledge base may contain data from user experiences. The objective of the knowledge base is to make the result more accurate and reliable.

Types of Data Mining architecture:

1. **No Coupling:** The no coupling data mining architecture retrieves data from particular data sources. It does not use the database for retrieving the data which is otherwise quite an efficient and accurate way to do the same. The no coupling architecture for data mining is poor and only used for performing very simple data mining processes.

2. **Loose Coupling:** In loose coupling architecture data mining system retrieves data from the database and stores the data in those systems. This mining is for memory-based data mining architecture.
3. **Semi-Tight Coupling:** It tends to use various advantageous features of the data warehouse systems. It includes sorting, indexing, and aggregation. In this architecture, an intermediate result can be stored in the database for better performance.
4. **Tight coupling:** In this architecture, a data warehouse is considered one of its most important components whose features are employed for performing data mining tasks. This architecture provides scalability, performance, and integrated information

Data Mining is considered as an interdisciplinary field. It includes a set of various disciplines such as statistics, database systems, machine learning, visualization and information sciences. Classification of the data mining system helps users to understand the system and match their requirements with such systems.

Data mining systems can be categorized according to various criteria, as follows:

1. **Classification according to the application adapted:** This involves domain-specific application. For example, the data mining systems can be tailored accordingly for telecommunications, finance, stock markets, e-mails and so on.
2. **Classification according to the type of techniques utilized:** This technique involves the degree of user interaction or the technique of data analysis involved. For example, machine learning, visualization, pattern recognition, neural networks, database-oriented or data-warehouse oriented techniques.
3. **Classification according to the types of knowledge mined:** This is based on functionalities such as characterization, association, discrimination and correlation, prediction etc.
4. **Classification according to types of databases mined:** A database system can be classified as a ‘type of data’ or ‘use of data’ model or ‘application of data’.

Associations and Correlations-

Association means that one variable provides information about another, but correlation means that two variables show an increasing or decreasing trend. Correlation means an association, but not causation.

Is association synonymous with correlation? (Association versus correlation)

It is important to understand that association is a non-technical term. Correlation is measured by a correlation coefficient, and requires that there is an association between two variables, but association does not imply correlation.

What is association with example?

An association is a "using" relationship between two or more objects in which the objects have their own lifetime and there is no owner. As an example, imagine the relationship between a doctor and a patient. A doctor can be associated with multiple patients.

What are the basic concepts of association and correlation?

Association analysis is used to discover co-occurrences or relationships between items in a dataset, while correlation analysis measures the strength of the relationship between two variables. In the subsequent sections, let's explore both techniques, their types, and various algorithms/methods to implement them.

Association Rule Mining:

Association rule mining is a technique used to uncover hidden relationships between variables in large datasets. It is a popular method in data mining and machine learning and has a wide range of applications in various fields, such as market basket analysis, customer segmentation, and fraud detection.

Association rule mining finds interesting associations and relationships among large sets of data items. This rule shows how frequently a itemset occurs in a transaction. A typical example is a Market Based Analysis.

Market Based Analysis is one of the key techniques used by large relations to show associations between items. It allows retailers to identify relationships between the items that people buy together frequently.

Given a set of transactions, we can find rules that will predict the occurrence of an item based on the occurrences of other items in the transaction.

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Before we start defining the rule, let us first see the basic definitions.

Support Count( ) – Frequency of occurrence of a itemset.

Here ({Milk, Bread, Diaper}) = 2

Frequent Itemset – An itemset whose support is greater than or equal to minsup threshold.

Association Rule – An implication expression of the form X -> Y, where X and Y are any 2 itemsets.

Example: {Milk, Diaper}->{Beer}

Rule Evaluation Metrics –

- **Support(s) –**

The number of transactions that include items in the {X} and {Y} parts of the rule as a percentage of the total number of transaction. It is a measure of how frequently the collection of items occur together as a percentage of all transactions.

- **Support = (X+Y) total –**

It is interpreted as fraction of transactions that contain both X and Y.

- **Confidence(c) –**

It is the ratio of the no of transactions that includes all items in {B} as well as the no of transactions that includes all items in {A} to the no of transactions that includes all items in {A}.

- **Conf(X=>Y) = Supp(X Y) Supp(X) –**

It measures how often each item in Y appears in transactions that contains items in X also.

- **Lift(l) –**

The lift of the rule X=>Y is the confidence of the rule divided by the expected confidence, assuming that the itemsets X and Y are independent of each other.The expected confidence is the confidence divided by the frequency of {Y}.

- **Lift(X=>Y) = Conf(X=>Y) Supp(Y) –**

Lift value near 1 indicates X and Y almost often appear together as expected, greater than 1 means they appear together more than expected and less than 1 means they appear less than expected. Greater lift values indicate stronger association.

Example – From the above table, {Milk, Diaper}=>{Beer}

s= ({Milk, Diaper, Beer}) / |T| = 2/5 = 0.4

c= (Milk, Diaper, Beer) (Milk, Diaper) = 2/3 = 0.67

l= Supp({Milk, Diaper, Beer}) Supp({Milk, Diaper})*Supp({Beer})

= 0.4/(0.6*0.6)

= 1.11

The Association rule is very useful in analyzing datasets. The data is collected using bar-code scanners in supermarkets. Such databases consists of a large number of transaction records which list all items bought by a customer on a single purchase. So the manager could know if certain groups of items are consistently purchased together and use this data for adjusting store layouts, cross-selling, promotions based on statistics.

Efficient and Scalable Frequent Item set Mining Methods –

1. Frequent item sets, also known as association rules, are a fundamental concept in association rule mining, which is a technique used in data mining to discover relationships between items in a dataset. The goal of association rule mining is to identify relationships between items in a dataset that occur frequently together.
2. A frequent item set is a set of items that occur together frequently in a dataset. The frequency of an item set is measured by the support count, which is the number of transactions or records in the dataset that contain the item set. For example, if a dataset contains 100 transactions and the item set {milk, bread} appears in 20 of those transactions, the support count for {milk, bread} is 20.
3. Association rule mining algorithms, such as Apriori or FP-Growth, are used to find frequent item sets and generate association rules. These algorithms work by iteratively generating candidate item sets and pruning those that do not meet the minimum support threshold. Once the frequent item sets are found, association rules can be generated by using the concept of confidence, which is the ratio of the number of transactions that contain the item set and the number of transactions that contain the antecedent (left-hand side) of the rule.
4. Frequent item sets and association rules can be used for a variety of tasks such as market basket analysis, cross-selling and recommendation systems. However, it should be noted that association rule mining can generate a large number of rules, many of which may be irrelevant or uninteresting. Therefore, it is important to use appropriate measures such as lift and conviction to evaluate the interestingness of the generated rules.

Association Mining searches for frequent items in the data set. In frequent mining usually, interesting associations and correlations between item sets in transactional and relational databases are found. In short, Frequent Mining shows which items appear together in a transaction or relationship.

Need of Association Mining: Frequent mining is the generation of association rules from a Transactional Dataset. If there are 2 items X and Y purchased frequently then it’s good to put them together in stores or provide some discount offer on one item on purchase of another item. This can really increase sales. For example, it is likely to find that if a customer buys **Milk** and **bread** he/she also buys **Butter**. So the association rule is **[‘milk]^[‘bread’]=>[‘butter’]**. So the seller can suggest the customer buy butter if he/she buys Milk and Bread.

Mining Various Kinds of Association Rules –

There are various types of association rules in data mining:-

- Multi-relational association rules
- Generalized association rules
- Quantitative association rules

- **1. Multi-relational association rules:** Multi-Relation Association Rules (MRAR) is a new class of association rules, different from original, simple, and even multi-relational association rules (usually extracted from multi-relational databases), each rule element consists of one entity but many a relationship. These relationships represent indirect relationships between entities.
- **2. Generalized association rules:** Generalized association rule extraction is a powerful tool for getting a rough idea of interesting patterns hidden in data. However, since patterns are extracted at each level of abstraction, the mined rule sets may be too large to be used effectively for decision-making. Therefore, in order to discover valuable and interesting knowledge, post-processing steps are often required. Generalized association rules should have categorical (nominal or discrete) properties on both the left and right sides of the rule.
- **3. Quantitative association rules:** Quantitative association rules is a special type of association rule. Unlike general association rules, where both left and right sides of the rule should be categorical (nominal or discrete) attributes, at least one attribute (left or right) of quantitative association rules must contain numeric attributes

Association Mining to Correlation Analysis –

What is correlation or association in sociology?

Association --- measures how closely related two variables are (i.e. whether they are dependent or independent).

Correlation --- measures in what way two variables are related (i.e. positive or negative).

Association refers to the discovery of co-occurrences or relationships between items in a dataset. On the other hand, correlation measures the strength of the relationship between two variables. It provides an insight into how the variables are related and how they affect each other.

Association analysis is most widely used to discover hidden patterns in large data sets. These hidden and uncovered relationships can be represented in the form of **association rules** or **sets of frequent items**. The role of identifying interesting associations in large databases is correlation analysis. There can be two types of these enthralling relationships: frequent itemsets or rules of the association. Frequent object sets are a collection of objects that mostly take place together. Association rules are the method of viewing fascinating relationships. The rules of association show that a close bond occurs between two or more objects.

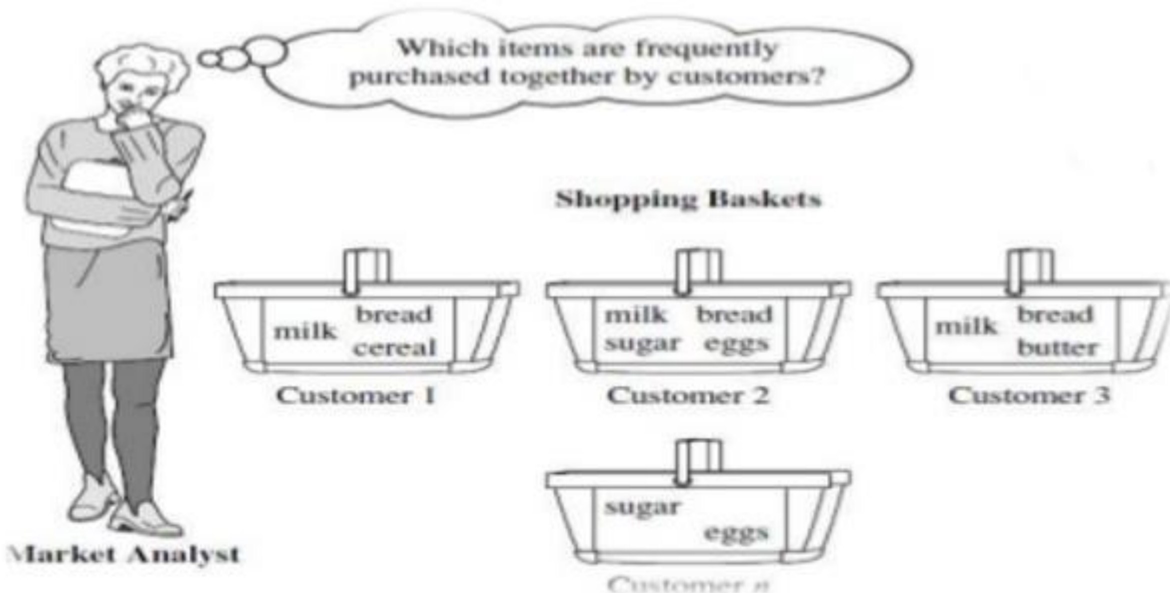


Fig: Market Basket Analysis

Market Basket Analysis

In transactional data, each case is connected with a set of objects. In principle, the list may contain all possible data items in the collection. For example - in a single market-basket analysis, goods with related items may be bought. However, only a small subset of all potential goods are present in a given set; only a small fraction of the items available for sale in the shop reflect the items in the market basket.

A common example of a regular pattern (item set) mining for association rules is market basket analysis. Business basket, the research analyzes the purchasing patterns of consumers by identifying correlations with the multiple items carried in their shopping baskets by customers.

An example of association rule - milk, bread

In a shop, if a shopkeeper sales milk then it is a probability to sell bread because a customer who is buying milk may also purchase bread. So it is showing that milk and bread are correlated with one another.

Constraint-Based Association Mining.

A data mining procedure can uncover thousands of rules from a given set of information, most of which end up being independent or tedious to the users. Users have a best sense of which "direction" of mining can lead to interesting patterns and the "form" of the patterns or rules they can like to discover.

Therefore, a good heuristic is to have the users defines such intuition or expectations as constraints to constraint the search space. This strategy is called constraint-based mining.

Constraint-based algorithms need constraints to decrease the search area in the frequent itemset generation step (the association rule generating step is exact to that of exhaustive algorithms).

The general constraint is the support minimum threshold. If a constraint is uncontrolled, its inclusion in the mining phase can support significant reduction of the exploration space because of the definition of a boundary inside the search space lattice, following which exploration is not needed.

The important of constraints is well-defined − they create only association rules that are appealing to users. The method is quite trivial and the rules space is decreased whereby remaining methods satisfy the constraints.

Constraint-based clustering discover clusters that satisfy user-defined preferences or constraints. It depends on the characteristics of the constraints, constraint-based clustering can adopt rather than different approaches.

The constraints can include the following which are as follows −

Knowledge type constraints − These define the type of knowledge to be mined, including association or correlation.

Data constraints − These define the set of task-relevant information such as Dimension/level constraints − These defines the desired dimensions (or attributes) of the information, or methods of the concept hierarchies, to be utilized in mining.

Interestingness constraints − These defines thresholds on numerical measures of rule interestingness, including support, confidence, and correlation.

Rule constraints − These defines the form of rules to be mined. Such constraints can be defined as metarules (rule templates), as the maximum or minimum number of predicates that can appear in the rule antecedent or consequent, or as relationships between attributes, attribute values, and/or aggregates.

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